

MINSEOK KANG

Computer Vision Engineer, ML/DL Researcher

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RESEARCH INTERESTS

Scene Understanding: Video Temporal Grounding, Video Scene Graph Generation

Vision-Language Multimodal Learning: Video Large Language Model

EDUCATION

Integrated M.S./Ph.D.

Mar. 2022 – Feb. 2027 (expected)

Yonsei University, Image and Video Pattern Recognition Lab.

Seoul, Korea

Electrical and Electronic Engineering, GPA 4.07/4.3

Bachelor of Science

Mar. 2017 – Feb. 2021

Korea University

Seoul, Korea

Electrical and Electronic Engineering, GPA 3.71/4.5 (Major 4.13/4.5)

Kyunggi High School

Mar. 2014 – Feb. 2017

Seoul, Korea

PUBLICATIONS

OTT-Vid: Optimal Transport Temporal Token Compression for Video Large Language Models

Minseok Kang, Minhyeok Lee, Jungho Lee, Minjung Kim, Donghyeong Kim,
Dayeon Lee, Heeseung Choi, Ig-Jae Kim, Sangyoun Lee

Under Review at NeurIPS 2026

Revisiting Weakly-Supervised Video Scene Graph Generation via Pair Affinity Learning

Minseok Kang, Minhyeok Lee, Minjung Kim, Jungho Lee,
Donghyeong Kim, Sungmin Woo, Inseok Jeon, Sangyoun Lee

Under Review at ECCV 2026

SwiftVGGT: A Scalable Visual Geometry Grounded Transformer for Large-Scale Scenes

Jungho Lee, Minhyeok Lee, Sunghun Yang, Minseok Kang, Sangyoun Lee

CVPR Findings 2026

Seen-to-Scene: Keep the Seen, Generate the Unseen for Video Outpainting

Inseok Jeon, Minhyeok Lee, Seunghoon Lee, Minseok Kang, Suhwan Cho, Sangyoun Lee

CVPR Findings 2026

Generalizing CLIP Prompts for Zero-Shot Anomaly Detection

Donghyeong Kim, Chaewon Park, Suhwan Cho, Hyeonjeong Lim, Minseok Kang,
Jungho Lee, Sangyoun Lee

Pattern Recognition 2026

Empower Words: DualGround for Structured Phrase and Sentence-Level Temporal Grounding

Minseok Kang, Minhyeok Lee, Minjung Kim, Donghyeong Kim, Sangyoun Lee

NeurIPS 2025

CMTM: Cross-Modal Token Modulation for Unsupervised Video Object Segmentation

Inseok Jeon, Suhwan Cho, Minhyeok Lee, Seunghoon Lee, Minseok Kang, Jungho Lee, Chaewon Park,
Donghyeong Kim, Sangyoun Lee

ICIP 2025

Multilevel Artificial Electronic Synaptic Device of Direct Grown Robust MoS₂ Based Memristor Array for In-Memory Deep Neural Network

Muhammad Naqi*, Minseok Kang*, Na Liu*, Taehwan Kim,
Seungho Baek, Arindam Bala, Changgyun Moon, Jongsun Park, Sunkook Kim

npj 2D Materials and Applications, 6, 53, Aug. 2022

* indicates equal contribution.

PROJECTS

Robust Detection of Abnormal and Hazardous Behaviors under Nighttime and Adverse Weather Conditions Jul. 2024 – Feb. 2027

Yonsei University | Funded by Electronics and Telecommunications Research Institute (ETRI)

- Project Leader / Deep Learning Researcher
- Construction of multi-sensor dataset integrating RGB, Depth, Thermal, IR, and LiDAR modalities
- Multi-sensor fusion for robust object and behavior recognition under low-visibility conditions
- Development of an efficient intelligent image processing algorithm resilient to complex degradations

Collaborative Perception and Intelligence Framework for Hyper-connected Interaction among Human and Intelligent Things Apr. 2024 – Dec. 2025

Korea Electronics Technology Institute (KETI)

- Deep Learning Researcher
- Development of an efficient skeleton-based action recognition model

Domain Generalization-Based Face Recognition Anti-Spoofing Model Aug. 2023 – Aug. 2024

Yonsei University | Funded by Samsung Electronics, Mobile eXperience (MX)

- Deep Learning Researcher
- Construction of a multi-device facial image database captured under diverse illumination and pose conditions
- Development of a domain generalized face anti-spoofing model robust across diverse camera domains

2D-3D Feature Correspondence Learning for Virtual Scene Reconstruction Jun. 2022 – Jun. 2023

Yonsei University | Funded by LG Electronics, Business Solutions (BS)

- Deep Learning Researcher
- Development of core technology for constructing a virtual 3D space by estimating 3D position and pose
- Development of stable 3D scene reconstruction from monocular 2D image

PATENTS

Method and Apparatus for Multi-domain Object Detection Based on Geometric Information 2026

KR Patent 10-2026-0026131

Multi-Domain Based Device and Method for Object Detection and Tracking 2025

KR Patent 10-2025-0024051

Electronic Device and Method for Detecting Forgery of Biometric Information 2024

KR Patent 10-2024-0123340

HONORS AND AWARDS

3rd Place, Perception Test Challenge 2025 — Task 3 2025

Temporal Action Localization & Temporal Sound Localization, Hosted by DeepMind (ICCV Workshop 2025)

Semester High Honors, School of Engineering, Korea University Spring 2019, Fall 2019, Spring 2020

Recognition for top academic performance

Top-Honor Scholarship, School of Electrical Engineering, Korea University 2019, 2020

Merit-based scholarship for outstanding academic achievement

SKILLS

Languages: Korean (Native), English (Professional working proficiency)

Programming (Advanced): Python, PyTorch, Linux OS, Bash/Shell scripting

Programming (Intermediate): C/C++, MATLAB, Verilog